If-Else Statements

Making decisions in C with if, else if, else and nested conditions

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What is an if statement?

An `if` statement runs a block of code only when a condition is **true**. In C, any non-zero value counts as true and `0` counts as false. `else if` adds more conditions that are checked one after another, and `else` runs when none of the conditions above it were true.

Syntax

```
if (condition1) {
    // runs when condition1 is true
} else if (condition2) {
    // runs when condition1 is false and condition2 is true
} else {
    // runs when none of the conditions are true
}
```

Operators used in conditions

Operator	Meaning	Example
`==`	equal to	`a == b`
`!=`	not equal to	`a != 0`
`< > <= >=`	comparison	`x >= 50`
`&&`	AND: both must be true	`a > 0 && b > 0`
`||`	OR: at least one must be true	`a == b || b == c`
`!`	NOT: reverses the result	`!flag`

Note the difference between `=` (assignment) and `==` (comparison). Writing `if (x = 5)` assigns 5 to x and is always true, which is a common bug.

EXAMPLE 1

Increase or decrease two numbers

Read two integers. If the first is smaller than the second, add 5 to the first; otherwise subtract 5 from the second. Print both final values.

CODE

```c
#include <stdio.h>

int main() {
    int num1, num2;

    printf("Enter the first number: ");
    scanf("%d", &num1);
    printf("Enter the second number: ");
    scanf("%d", &num2);

    if (num1 < num2) {
        num1 += 5;          /* increase num1 by 5 */
    } else {
        num2 -= 5;          /* decrease num2 by 5 */
    }

    printf("First number: %d\n", num1);
    printf("Second number: %d\n", num2);
    return 0;
}
```

SAMPLE RUN

```
Enter the first number: 3
Enter the second number: 8
First number: 8
Second number: 8
```

EXAMPLE 2

Changing a value by range (else if chain)

Read an integer. If it is less than 50, add 10. If it is between 50 and 100 (inclusive), subtract 5. If it is greater than 100, add 3. Print the new value.

CODE

```c
#include <stdio.h>

int main() {
    int number;

    printf("Enter a number: ");
    scanf("%d", &number);

    if (number < 50) {
        number += 10;
    } else if (number <= 100) {     /* here number is already >= 50 */
        number -= 5;
    } else {
        number += 3;
    }

    printf("New value: %d\n", number);
    return 0;
}
```

SAMPLE RUN

```
Enter a number: 75
New value: 70
```

Note. The second condition only needs `number <= 100`, because the program reaches it only when `number < 50` was false.

EXAMPLE 3

Adjusting a temperature

Read a temperature. Below 10 degrees, increase it by 1. Between 10 and 30 (inclusive), leave it unchanged. Above 30, decrease it by 2.

CODE

```c
#include <stdio.h>

int main() {
    int temperature;

    printf("Enter the temperature: ");
    scanf("%d", &temperature);

    if (temperature < 10) {
        temperature++;              /* increase by 1 */
    } else if (temperature > 30) {
        temperature -= 2;           /* decrease by 2 */
    }
    /* 10..30: no change, so no branch is needed */

    printf("Temperature: %d\n", temperature);
    return 0;
}
```

SAMPLE RUN

```
Enter the temperature: 34
Temperature: 32
```

EXAMPLE 4

Leap year check

Read a year and decide whether it is a leap year. A year is a leap year if it is divisible by 4 but not by 100, or if it is divisible by 400.

CODE

```c
#include <stdio.h>

int main() {
    int year;

    printf("Enter a year: ");
    scanf("%d", &year);

    if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0) {
        printf("%d is a leap year.\n", year);
    } else {
        printf("%d is not a leap year.\n", year);
    }
    return 0;
}
```

SAMPLE RUN

```
Enter a year: 2024
2024 is a leap year.
```

Note. 1900 is not a leap year (divisible by 100 but not 400), while 2000 is (divisible by 400).

EXAMPLE 5

Largest of three numbers and equality check

Read three integers. Find the largest one, then report whether all three are equal, two of them are equal, or none are equal.

CODE

```c
#include <stdio.h>

int main() {
    int a, b, c;

    printf("Enter three numbers (separated by spaces): ");
    scanf("%d %d %d", &a, &b, &c);

    /* Find the largest number */
    if (a >= b && a >= c) {
        printf("Largest number: %d\n", a);
    } else if (b >= a && b >= c) {
        printf("Largest number: %d\n", b);
    } else {
        printf("Largest number: %d\n", c);
    }

    /* Check equality */
    if (a == b && b == c) {
        printf("All three numbers are equal.\n");
    } else if (a == b || a == c || b == c) {
        printf("Two of the numbers are equal.\n");
    } else {
        printf("No two numbers are equal.\n");
    }
    return 0;
}
```

SAMPLE RUN

```
Enter three numbers (separated by spaces): 7 12 7
Largest number: 12
Two of the numbers are equal.
```

EXAMPLE 6

Triangle type finder (nested if)

Read three side lengths. Check that they are positive and can form a triangle (each pair of sides must be longer than the third side). If so, classify the triangle as equilateral, isosceles or scalene.

CODE

```c
#include <stdio.h>

int main() {
    float a, b, c;

    printf("Enter the three side lengths: ");
    scanf("%f %f %f", &a, &b, &c);

    if (a <= 0 || b <= 0 || c <= 0) {
        printf("Side lengths must be positive.\n");
    } else if (a + b > c && a + c > b && b + c > a) {
        if (a == b && b == c) {
            printf("This is an equilateral triangle.\n");
        } else if (a == b || b == c || a == c) {
            printf("This is an isosceles triangle.\n");
        } else {
            printf("This is a scalene triangle.\n");
        }
    } else {
        printf("These lengths cannot form a triangle.\n");
    }
    return 0;
}
```

SAMPLE RUN

```
Enter the three side lengths: 5 5 8
This is an isosceles triangle.
```

EXAMPLE 7

Area of a triangle by type

Ask the user to choose a triangle type. For a right triangle, read the base and height. For an equilateral triangle, read one side. Calculate and print the area.

> Right triangle: Area = (base x height) / 2
> Equilateral triangle: Area = (side^2 x √3) / 4, where √3 ≈ 1.732

CODE

```c
#include <stdio.h>

int main() {
    int type;
    float base, height, side, area;

    printf("Choose a triangle type:\n");
    printf("1. Right triangle\n");
    printf("2. Equilateral triangle\n");
    scanf("%d", &type);

    if (type == 1) {
        printf("Enter the base: ");
        scanf("%f", &base);
        printf("Enter the height: ");
        scanf("%f", &height);
        area = (base * height) / 2;
        printf("Area of the right triangle: %.2f\n", area);
    } else if (type == 2) {
        printf("Enter the side length: ");
        scanf("%f", &side);
        area = (side * side * 1.732) / 4;   /* 1.732 is approx. sqrt(3) */
        printf("Area of the equilateral triangle: %.2f\n", area);
    } else {
        printf("Invalid choice!\n");
    }
    return 0;
}
```

SAMPLE RUN

```
Choose a triangle type:
1. Right triangle
2. Equilateral triangle
2
Enter the side length: 14
Area of the equilateral triangle: 84.87
```

EXAMPLE 8

Area of a geometric shape

Let the user choose a triangle, rectangle or circle, read the required dimensions and print the area. The constant PI is defined with `#define`.

CODE

```c
#include <stdio.h>
#define PI 3.14159

int main() {
    int choice;
    float area, base, height, length, width, radius;

    printf("Choose a shape:\n");
    printf("1. Triangle\n");
    printf("2. Rectangle\n");
    printf("3. Circle\n");
    scanf("%d", &choice);

    if (choice == 1) {
        printf("Enter the base: ");
        scanf("%f", &base);
        printf("Enter the height: ");
        scanf("%f", &height);
        area = (base * height) / 2;
        printf("Area of the triangle: %.2f\n", area);
    } else if (choice == 2) {
        printf("Enter the length: ");
        scanf("%f", &length);
        printf("Enter the width: ");
        scanf("%f", &width);
        area = length * width;
        printf("Area of the rectangle: %.2f\n", area);
    } else if (choice == 3) {
        printf("Enter the radius: ");
        scanf("%f", &radius);
        area = PI * radius * radius;
        printf("Area of the circle: %.2f\n", area);
    } else {
        printf("Invalid choice!\n");
    }
    return 0;
}
```

SAMPLE RUN

```
Choose a shape:
1. Triangle
2. Rectangle
3. Circle
3
Enter the radius: 7
Area of the circle: 153.94
```

EXAMPLE 9

Exam evaluation (nested if-else)

Read a student's exam score. Print "Passed" for 50 and above, "Failed" below 50. Then add a second level of feedback: Excellent / Good / Average for passing scores, and a study message for failing scores.

CODE

```c
#include <stdio.h>

int main() {
    int score;

    printf("Enter the student's score: ");
    scanf("%d", &score);

    if (score >= 50) {
        printf("Passed\n");

        if (score >= 85) {
            printf("Congratulations! Grade: Excellent\n");
        } else if (score >= 70) {
            printf("Grade: Good\n");
        } else {
            printf("Grade: Average\n");
        }
    } else {
        printf("Failed\n");

        if (score >= 30) {
            printf("You need to study more!\n");
        } else {
            printf("You should start studying from the beginning.\n");
        }
    }
    return 0;
}
```

SAMPLE RUN

```
Enter the student's score: 80
Passed
Grade: Good
```